

BRAIN TUMOR SEGMENTATION USING CLASSICAL IMAGE PROCESSING TECHNIQUES

PRESENTED BY:

**MOHAMED HOSSAM
RANIM HISHAM**

INTRODUCTION

Goal: Detect and segment brain tumors in MRI images using classical image processing.

Dataset: Brain MRI dataset (Tumor vs No Tumor)

TOOLS: Python, OpenCV, Scikit-learn, Matplotlib

PIPELINE

Tumor Segmentation Pipeline

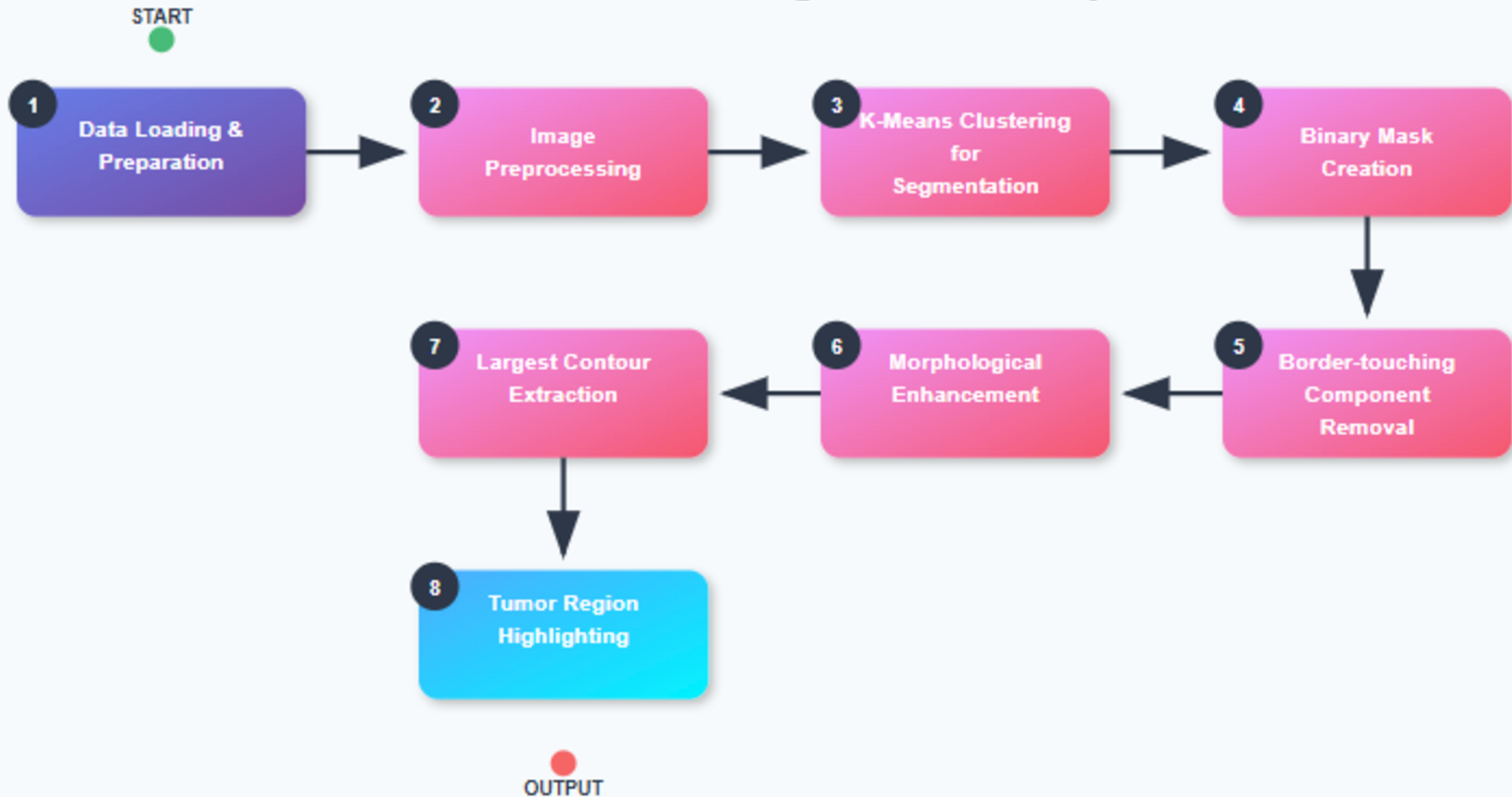
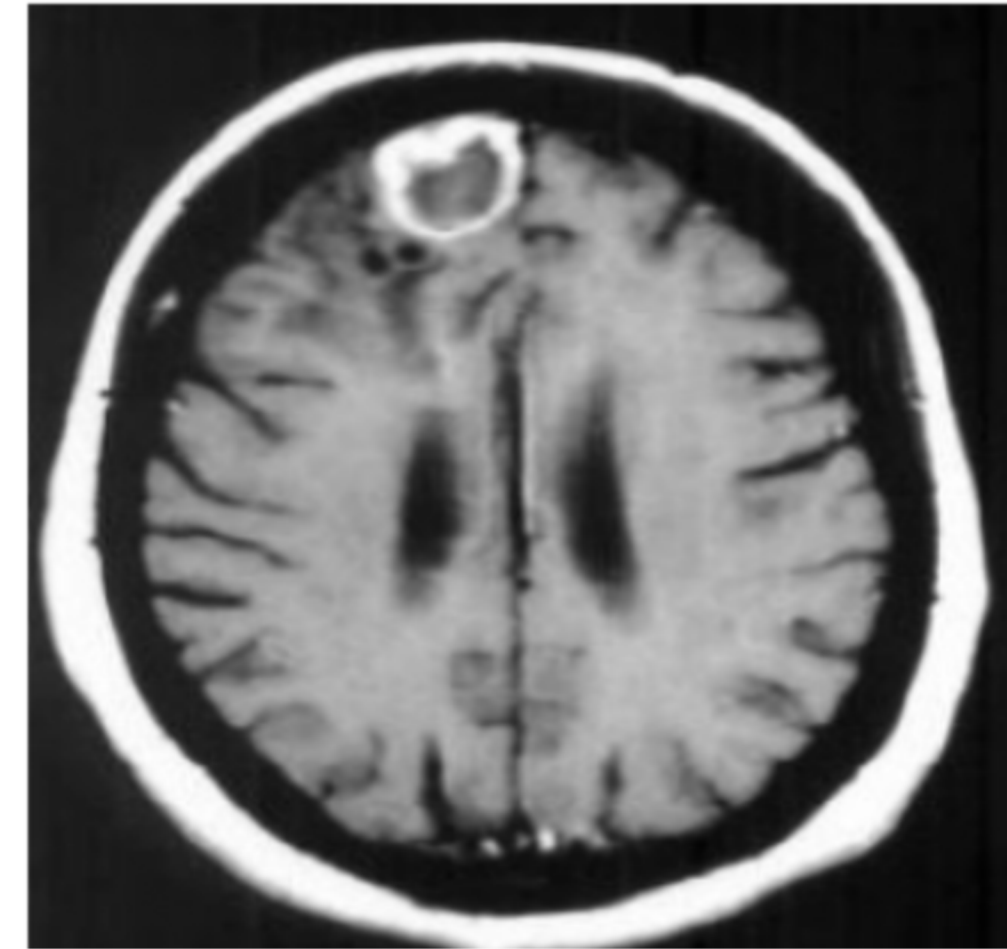


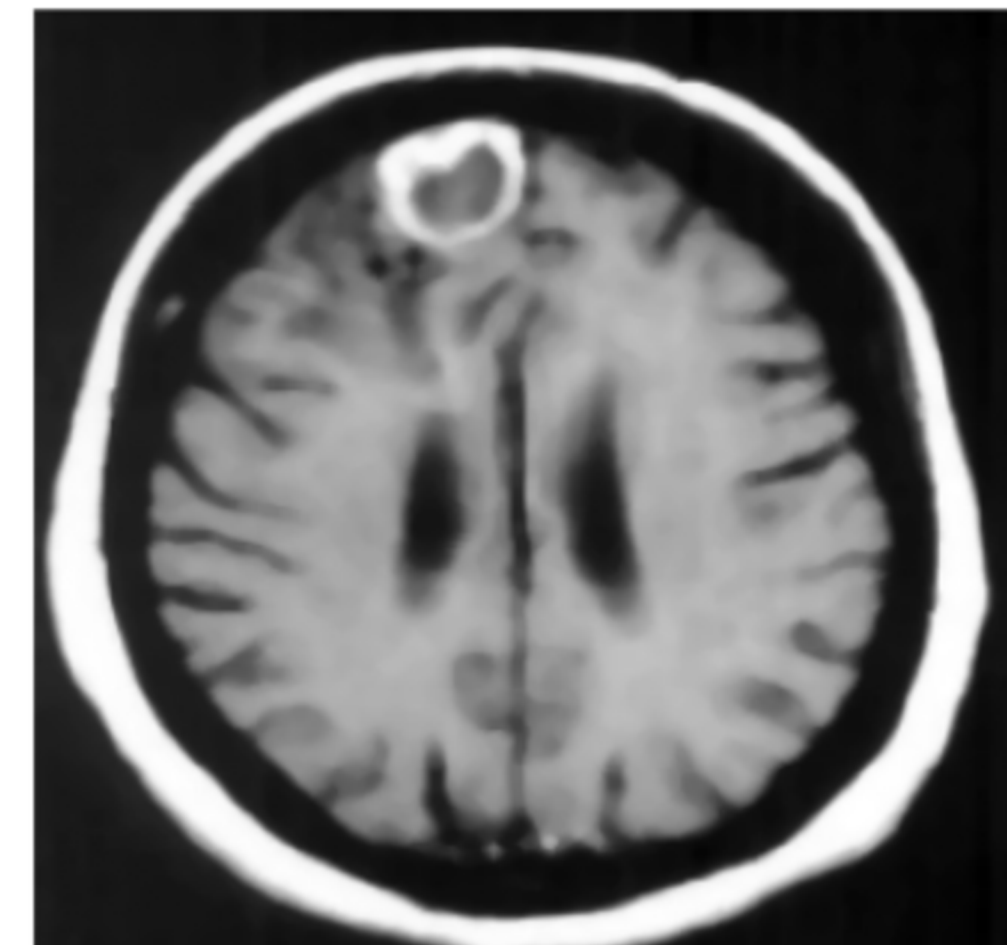
Image Preprocessing

- Convert images to grayscale (if not already).
- Resize all images to a fixed dimension (256x256) for consistency.
- Apply Median Filtering to reduce noise while preserving edges.
- This step enhances cluster performance by smoothing intensity variations.

Original Scans #4
(Tumor)



Preprocessed Images #4
(Tumor)



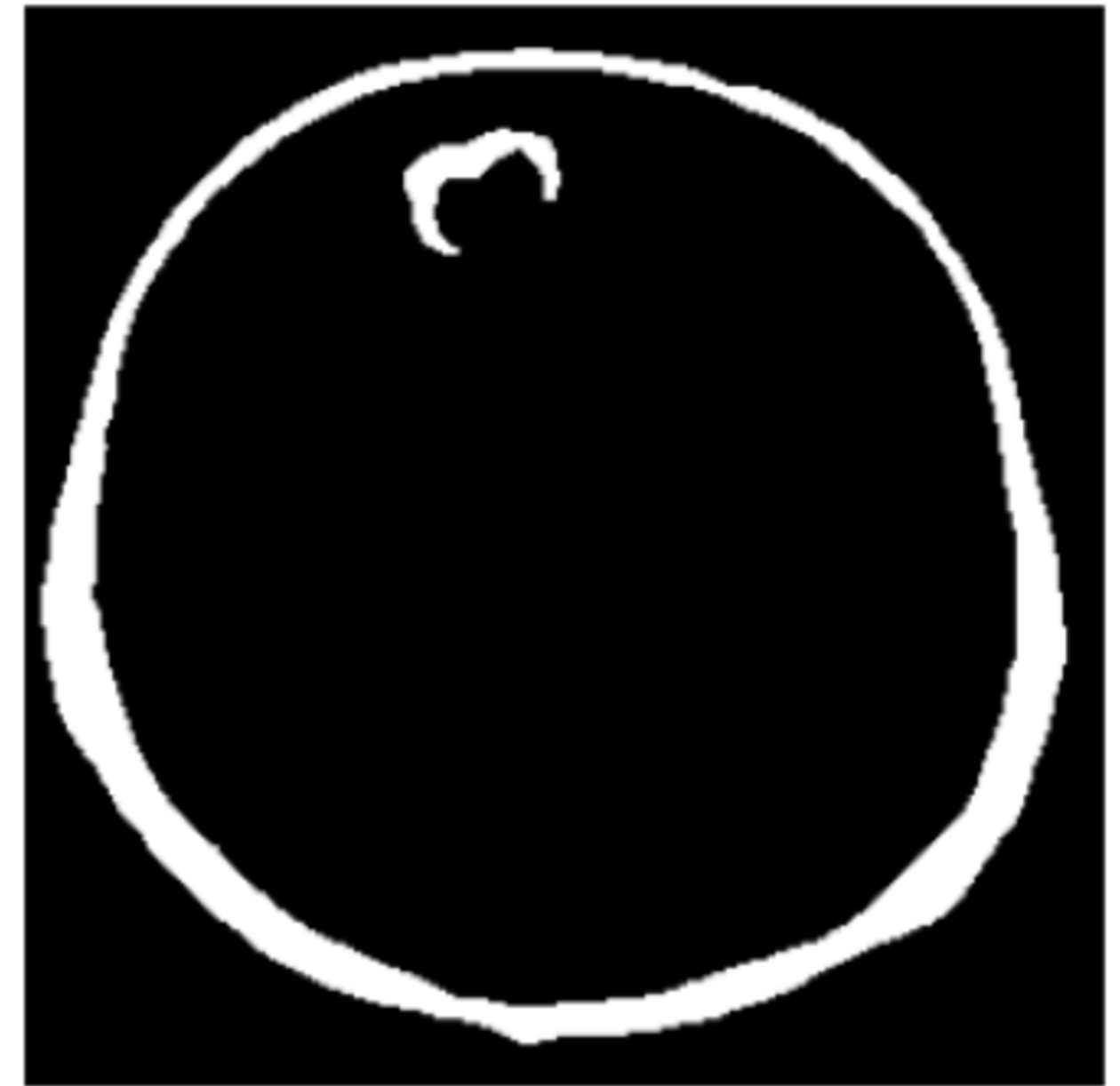
K-Means Clustering for Segmentation

- Apply K-Means clustering on pixel intensities (flattened grayscale image).
- Group pixels into clusters (e.g., 4 clusters) based on intensity similarity.

Binary Mask Creation

- Generate a binary mask from the selected intensity cluster
- Pixels from the suspected tumor cluster $\rightarrow$ 255 (white)
All other pixels $\rightarrow$ 0 (black)
- This serves as an initial segmentation, not a final tumor mask.

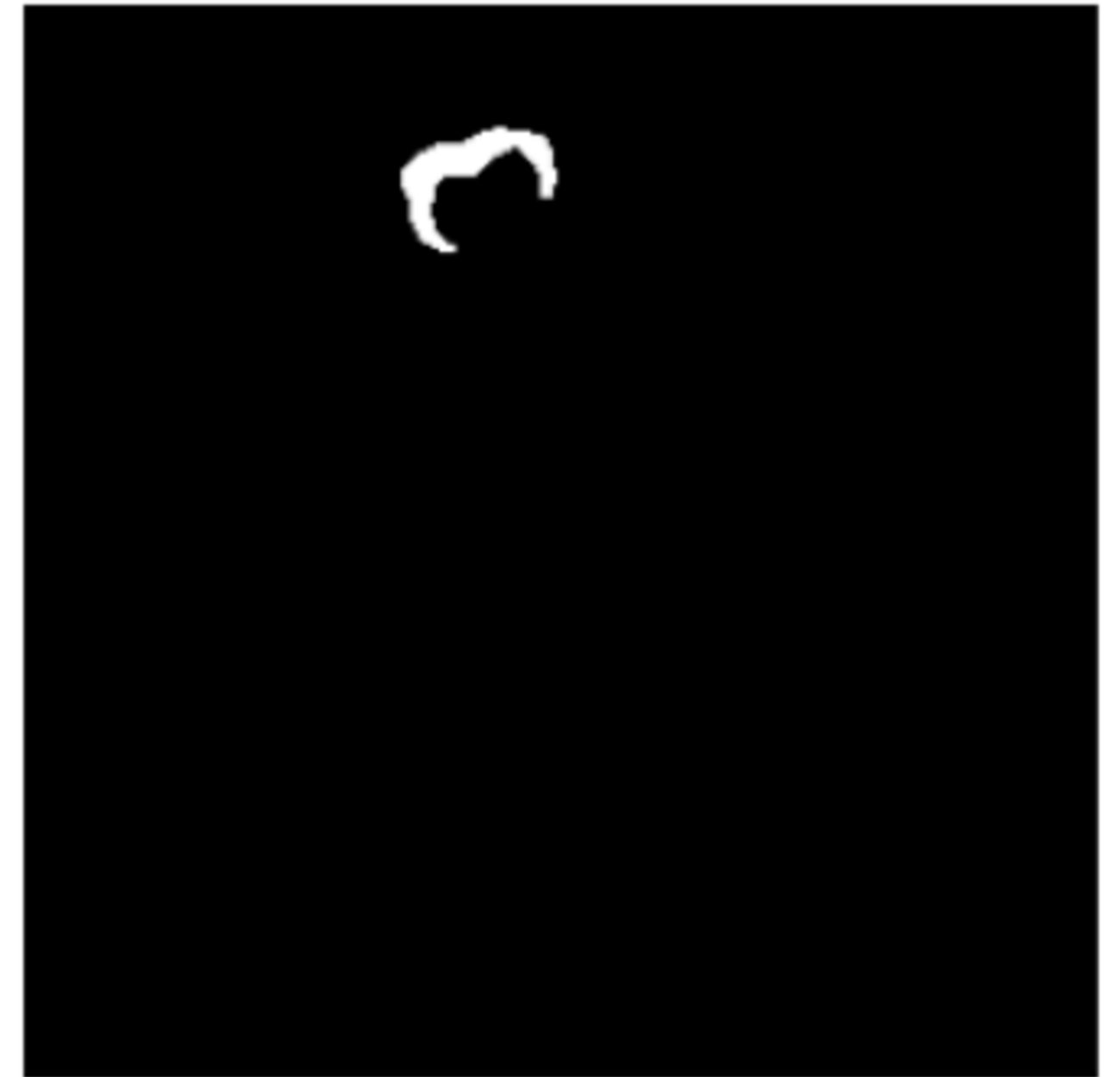
Raw Tumor Masks #4
(Tumor)



Border-touching Component Removal

- Detect all connected components in the binary mask.
- Remove components that touch or are too close to image borders (e.g., ≤ 10 pixels).
- This removes potential non-tumor regions.

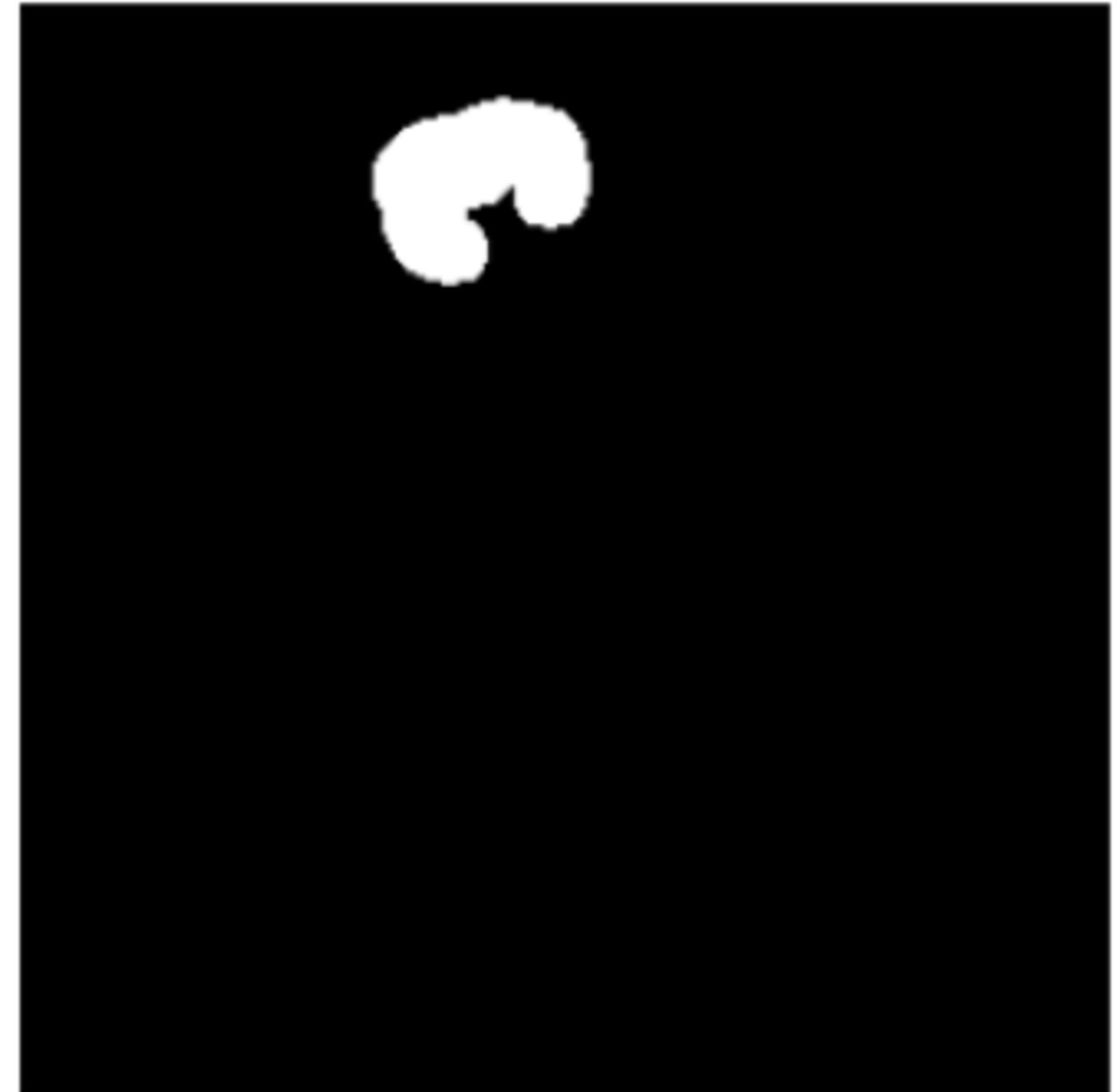
Cleaned Tumor Masks (No Border) #4
(Tumor)



Morphological Enhancement & Largest Contour Extraction

- Apply elliptical dilation to merge fragmented regions.
- Smooth boundaries and fill small holes in the mask.
- Detect external contours from the dilated mask.
- Select and fill the largest contour (likely tumor).
- Produce a clean, focused tumor segmentation mask.

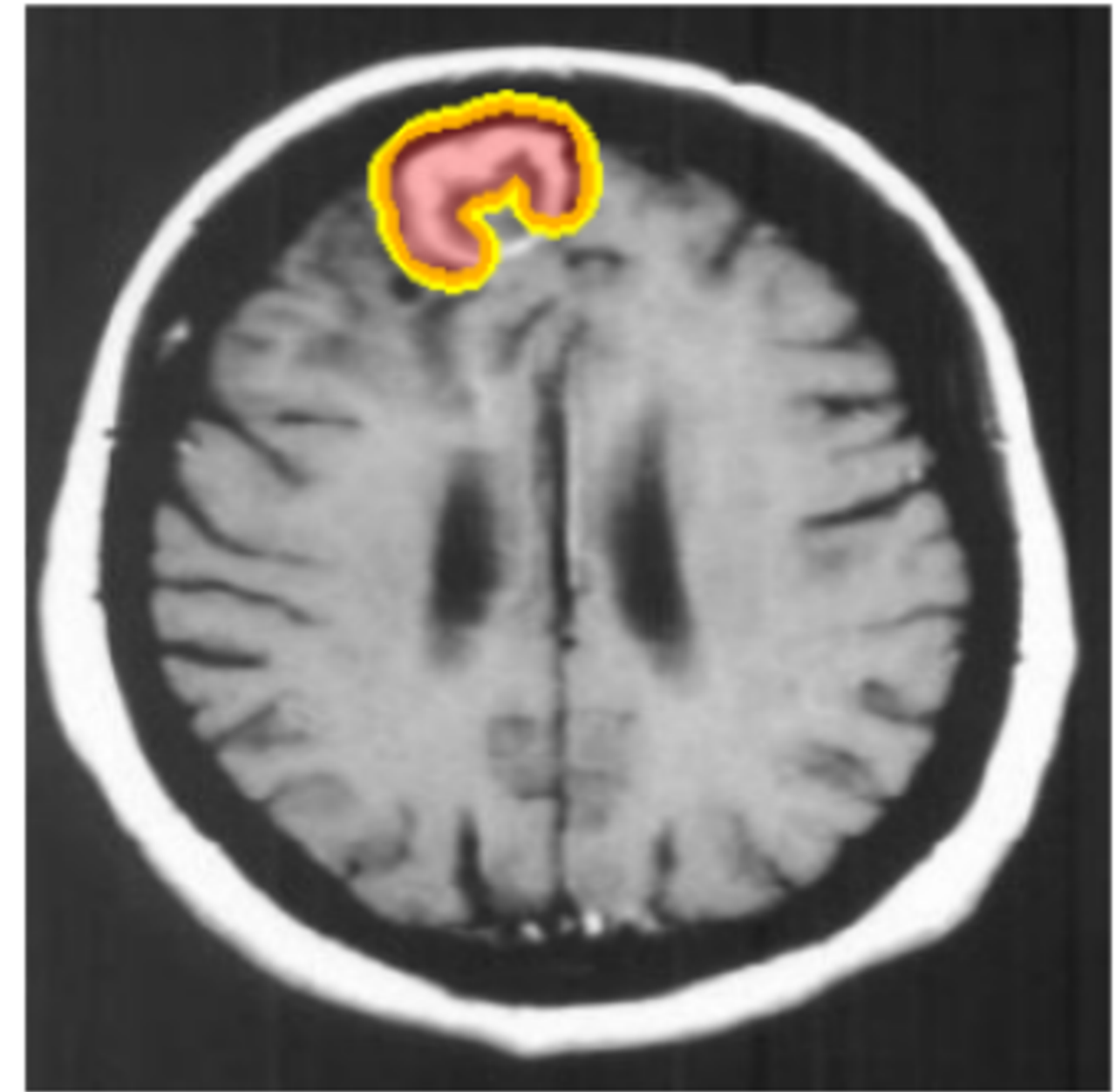
Largest Tumor Contour #4
(Tumor)



Tumor Region Highlighting

- Overlay the largest contour mask onto the original image.
- Highlight detected tumor region (e.g., with red color or transparency).
- Provides a visual representation of where the tumor lies in the MRI scan.

Tumor Overlay on Original #4
(Tumor)



Thank you!